DD-Mamba: A Forecast-Decomposable Dual-Domain State-Space Framework for Multivariate Time-Series Forecasting

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Abstract

Long-term multivariate forecasting requires a model to represent temporal dynamics, periodic structure, and cross-variable dependence without assuming that every dataset benefits from the same capacity. Existing dual-domain models usually fuse these sources internally, making their individual effects difficult to assess. We present DD-Mamba, a forecast-decomposable state-space framework in which a decomposition-based temporal branch and a learned spectral branch each produce a complete forecast and are combined by a convex gate. A switchable bidirectional variate mixer provides cross-channel capacity, while a linear temporal pathway and zero-initialized corrections give the model a simple starting map. The design supports controlled component removal rather than inferring component value from a single end-to-end score. On six benchmarks rerun in one pipeline against seven baselines, DD-Mamba obtains the lowest average MSE on five datasets; seed-matched tests with Benjamini–Hochberg correction support improvements over the strongest in-pipeline competitor on four, while Weather is tied and

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PatchTST is better on ETTm1. Across 83 component comparisons, none of the 29 placement effects survives multiplicity correction at three seeds. Eight corrected branch-removal effects support the temporal branch, chiefly on ETTm data. The only corrected frequency-side effect occurs for Solar at horizon 96, but because that dataset's normalization switch was selected with test feedback, we treat it as exploratory rather than confirmatory. Training-set channel correlation and dispersion statistics provide descriptive hypotheses for capacity placement, not a validated selector. The results therefore support a component-wise analysis framework and bounded dataset-specific findings, not universal superiority or a general rule for model selection.
Keywords: Time series forecasting, State-space models, Mamba, Frequency domain, Multivariate analysis

1. Introduction

Long-term forecasting of multivariate time series underpins applications from power-grid load planning to traffic management and weather-dependent operations. The field has cycled through architectural paradigms at a remarkable pace: Transformer variants with efficient attention [1, 2, 3], patch-based attention [4], inverted variate-token attention [5], and, most recently, selective state-space models (SSMs) built on Mamba [6, 7, 8], which offer linear-time sequence modeling with input-dependent dynamics.

Combining time- and frequency-domain representations is itself an established direction: TF4TF [9] models multi-semantic time–frequency information for long-term forecasting, and recent work pairs time–frequency or decomposition-based modeling with Mamba backbones [10, 8]. We therefore do *not* position dual-domain forecasting or a time–frequency Mamba as our contribution. What remains under-examined is *when* each such component actually pays: existing dual-domain models fuse representations inside the network, so the time view, the spectral view, normalization, and cross-variate capacity are entangled and cannot be individually switched off and measured. Our contribution is a *forecast-decomposable* framework that makes these choices separable and testable, together with the seed-paired, multiplicity-controlled evidence of where each one helps.

Three empirical gaps motivate this design. **Gap 1: deep models discard strong linear maps.** A per-component linear projection from the look-back window to the horizon (DLinear [11]) and a tiny complex-linear interpolation of the low-passed spectrum (FITS [12]) can be competitive with much larger models on standard benchmarks. Without such a direct pathway, a deep architecture must learn any linear behavior implicitly, which may be inefficient on small datasets. **Gap 2: the useful inductive bias differs by domain, but is rarely separable.** Local trends and regime changes are naturally expressed in the time domain, multi-scale seasonality in the frequency domain. Single-view models [13] expose only one, and dual-domain models such as TF4TF [9] fuse both *inside* the network, so neither view is a separately removable forecast path whose marginal value can be measured. **Gap 3: cross-variate capacity can help or hurt.** Cross-variate mixing is effective on several high-dimensional benchmarks [5, 7, 14], but the property it exploits — how strongly channels correlate — varies threefold across the standard benchmarks (Table 1), motivating an explicit switch rather than a universal setting.

We answer these three gaps with DD-Mamba (Fig. 1), mapping each to one structural remedy: *(i)* a DLinear-style time backbone with zero-initialized nonlinear corrections; *(ii)* parallel time and frequency forecasts, including an optional dormant FITS-style spectral map, fused by a convex gate; and *(iii)* a bidirectional variate mixer switched on or off per dataset. Seed-matched ablations test these choices below; effect sizes and confidence intervals are reported for every comparison, and a finding is considered statistically supported only when it survives the stated multiplicity correction.

Concretely, the time branch pairs a DLinear-style pathway [11] with a zero-initialized nonlinear correction and the frequency branch pairs learned filtering with an optional zero-initialized FITS-style spectral pathway [12], so the model starts as a linear map of the instance-normalized window (the RLinear-class structure [15, 16]) and the branch forecasts form a learned convex combination. Because each branch emits a full forecast, we can remove it and measure the effect with seed-paired tests under Benjamini–Hochberg (BH) correction. Two patterns emerge. Component-placement effects — the variate mixer helping Weather while hurting ETTh1, normalization helping Weather while hurting Solar — are visible raw but, at three seeds, *none* survive BH on the nine original benchmarks, so we report them as *exploratory*. Branch removal is sharper: of 54 tests, nine meet the BH threshold, eight of them on the *time* branch (ETTm1, ETTm2, Electricity, Exchange); the ninth, the frequency-side Solar cell at H=96, is itself exploratory because Solar's normalization switch was selected after test-set inspection. On PEMS04, by contrast, six of eleven component effects survive, led by cross-variate mixing. The clean evidence therefore supports the temporal pathway and the variate mixer, but does not establish a general dual-domain advantage.

Contributions. **(1)** A *forecast-decomposable* architecture in which temporal and spectral paths emit complete predictions and the normalization, variate mixer, and branch pathways can be removed without redefining the forecasting task. **(2)** A component-level evaluation protocol that reports seed-matched effect sizes and confidence intervals and controls the false discovery rate across 94 comparisons, distinguishing supported effects — the eight temporal-branch removals and the six PEMS04 components — from exploratory placement and frequency-side findings. **(3)** A data-centric analysis relating the measured effects to training-set cross-channel correlation, horizon, and dispersion. These statistics are used to formulate testable hypotheses, not presented as a validated selection rule.

Organization. Section 2 reviews the related literature and the resulting research gap and Section 3 the state-space background. Section 4 details the architecture, Section 5 the main comparison and ablations, and Section 6 the component-selection guidance and threats to validity.

2. Related Work

2.1. Time Series Analysis Tasks

Deep models now serve the full spectrum of time series mining tasks — forecasting, classification, anomaly detection, and imputation — often through shared backbones such as TimesNet [17]. Long-term multivariate *forecasting*, our setting, is the most benchmark-standardized of these: fixed look-back, multi-horizon evaluation, and shared splits [1, 5] make it a suitable arena for controlled architectural evidence. Backbone ideas transfer across tasks, but the component effects measured here are specific to forecasting.

2.2. Temporal-Dependency Models

One line models dependencies along the time axis only. Transformer variants adapt attention to long sequences via sparsity, decomposition, and frequency enhancement [1, 2, 3], patching [4], or explicit de-/re-stationarization [18]. A second family shows that *linear* maps remain competitive: DLinear/NLinear [11], RLinear with reversible instance normalization [15, 16], the MLP-based TiDE [19] and TimeMixer [20], and the residual-stacked N-BEATS/N-HiTS [21, 22], our closest residual-forecasting precedents. A third works in the spectrum: FreTS [13] learns frequency-domain MLPs and FITS [12] forecasts with one complex-linear interpolation of the low-passed spectrum. A fourth, directly adjacent to us, combines *both* views: TF4TF [9] mines multi-semantic time–frequency information for long-term forecasting. Such dual-domain models fuse the two views inside the network, so — unlike the framework here — neither view is a separately removable forecast path. Decomposition-based designs (STL [23], SCINet [24], and the seasonal-trend-dispersion view of STD [25]) make component structure explicit; RevIN, used here as a per-dataset switch, normalizes windows reversibly but neither extracts nor forecasts dispersion, and we claim no STD-style decomposition.

2.3. Cross-Dimensional (Spatial–Temporal) Models

A second line treats the variate dimension as a first-class axis. Crossformer [14] embeds the series into a two-dimensional patch grid and attends over *both* the temporal and the cross-variate (spatial) dimension. iTransformer [5] inverts the axes and attends over variate tokens only; SOFTS [26] reaches linear-complexity channel interaction through a centralized core, and ModernTCN [27] interleaves temporal and cross-variable convolutions. Selective state-space models bring linear-time scans to the same axes: Mamba [6, 28] over time, and S-Mamba [7], TimeMachine [29], and FMamba [30] over variate tokens, with recent variants adding multi-scale processing (ms-Mamba [8]) and series decomposition (DecMamba [10]). G-Mamba [31] further combines graph-conditioned state-space dynamics with explicit spatial relations, and TimePro [32] builds variable- and time-aware hyper-states to address multi-delay cross-variable effects. DD-Mamba sits in this lineage — causal Mamba over time and bidirectional Mamba over variates and frequency bins — but treats cross-variable capacity as a switchable component whose measured value can be analyzed.

2.4. Research Gap

Dual-domain forecasting [9], time–frequency and decomposition Mamba hybrids [10, 8], and cross-dimensional attention [14, 5] are all, by now, well-populated spaces; a new model in any of them is unlikely to be a contribution on its own. What these lines share is a *methodological* limitation rather than an architectural one. (i) Representations are fused inside the network, so the marginal value of the time view, the spectral view, or a decomposition step is never separately measured. (ii) Cross-dimensional models such as Crossformer apply the same spatial–temporal capacity to every dataset, although channel coupling varies enormously across benchmarks (mean absolute cross-channel Pearson correlation ranges from 0.31 on ETTh1 to 0.91 on Solar; Table 1). (iii) Component-level comparisons, when reported at all, typically rest on a handful of seeds and uncorrected p-values, so it is unclear which claimed effects would survive multiplicity control. The gap we address is therefore *measurement*. The closest works in spirit are TimeRecipe [33] and CombinationTS [34], which decompose forecasters into interchangeable modules and benchmark module-level effectiveness *across* many architectures at large scale. We do not attempt that cross-architecture generality: we audit components *within* a single forecast-decomposable architecture — independently removable time/frequency forecasts, a switchable variate mixer, and a 27-cell branch-ablation matrix — using per-seed paired tests with Benjamini–Hochberg control and explicitly labelling any switch chosen with test feedback as exploratory. A formal estimand, multiple backbones, or successful held-out component selection would be needed to raise this audit to the level of those frameworks. The result is bounded empirical findings on the studied datasets, not a validated rule for predicting component value on unseen data.

3. Preliminaries

3.1. Problem Formulation

Given a look-back window $\mathbf{X} \in \mathbb{R}^{L \times C}$ of L steps and C variates, long-term multivariate forecasting predicts the next H steps $\mathbf{Y} \in \mathbb{R}^{H \times C}$, with H typically in $\{96, 192, 336, 720\}$. Following RevIN [16], we standardize each window per variate, $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$, using the window's own per-variate mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$, and de-normalize the final forecast by the inverse affine map. Instance normalization is treated as a per-dataset switch rather than a universal default; Section 5.3 shows why. A model is a map $f_\theta : \tilde{\mathbf{X}} \mapsto \mathbf{Y}$ trained to minimize the mean squared error over the horizon.

3.2. State Space Models

A continuous linear state-space model maps an input signal $u(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through an N-dimensional latent state $\mathbf{h}(t) \in \mathbb{R}^N$,

$$\mathbf{h}'(t) = \mathbf{A}\,\mathbf{h}(t) + \mathbf{B}\,u(t), \qquad y(t) = \mathbf{C}^\top \mathbf{h}(t), \tag{1}$$

with $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B}, \mathbf{C} \in \mathbb{R}^N$. To operate on a discrete sequence, the system is discretized with a step size $\mathbf{\Delta}$ under a zero-order hold, which yields the discrete parameters

$$\bar{\mathbf{A}} = \exp(\mathbf{\Delta A}), \qquad \bar{\mathbf{B}} = (\mathbf{\Delta A})^{-1}(\exp(\mathbf{\Delta A}) - \mathbf{I})\,\mathbf{\Delta B} \;\approx\; \mathbf{\Delta B}, \tag{2}$$

and turns Eq. (1) into the linear recurrence $\mathbf{h}_k = \bar{\mathbf{A}}\,\mathbf{h}_{k-1} + \bar{\mathbf{B}}\,u_k$, $y_k = \mathbf{C}^\top \mathbf{h}_k$. Classical structured SSMs keep $(\bar{\mathbf{A}}, \bar{\mathbf{B}}, \mathbf{C})$ fixed across the sequence, so the recurrence is linear time-invariant and can be evaluated as a global convolution. Mamba [6] makes the model *selective*: the parameters $(\mathbf{\Delta}_k, \mathbf{B}_k, \mathbf{C}_k)$ become functions of the input u_k, so the state transition adapts at each step and the model can selectively remember or forget context. Selectivity breaks the convolutional form, but a hardware-aware parallel scan keeps the cost linear in sequence length. DD-Mamba uses this selective recurrence in two roles — a causal scan over time and, optionally, a bidirectional scan over variates — detailed next.

4. Method

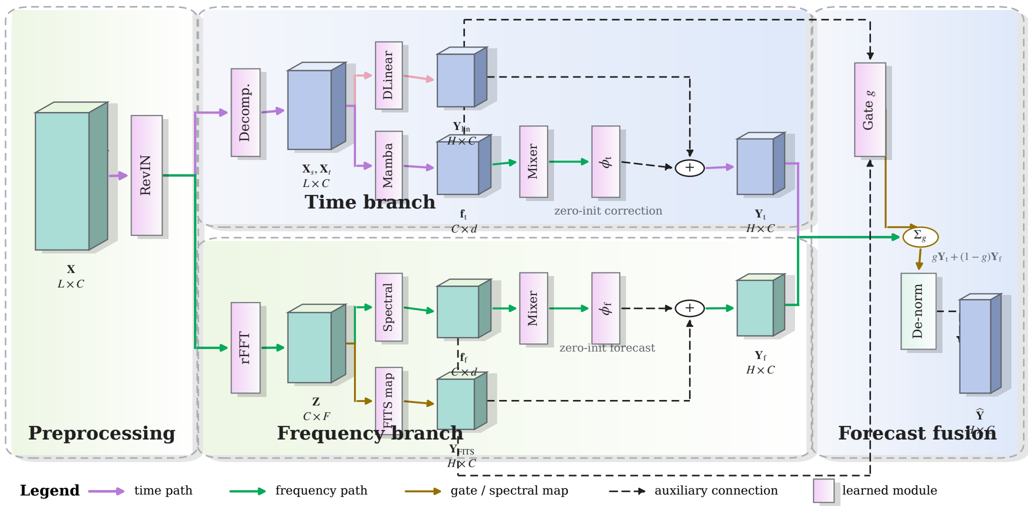

Figure 1: DD-Mamba as a forecast-decomposable architecture. The normalized input is routed to two explicit prediction paths. The time branch combines a DLinear anchor with a causal-Mamba correction; the frequency branch combines a complex spectral encoder with an optional FITS-style linear spectral map. Each branch produces a complete forecast, $\mathbf{Y}_\mathrm{t}$ or $\mathbf{Y}_\mathrm{f}$, and branch features parameterize the convex gate g. Shadowed cuboids denote intermediate tensors, vertical lavender bars denote learned operators, purple and green arrows trace the time- and frequency-domain paths, gold arrows mark spectral/gating operations, and black dashed arrows denote auxiliary or residual connections. Zero initialization and dataset-level switches are stated beside their associated operators. The final forecast is $g\mathbf{Y}_\mathrm{t} + (1-g)\mathbf{Y}_\mathrm{f}$, followed by inverse instance normalization.

4.1. Time Branch: Decomposition, Linear Backbone, Selective Scan

The time branch decomposes $\tilde{\mathbf{X}}$ with a moving average into trend $\mathbf{X}_t$ and seasonal $\mathbf{X}_s$ components [2, 11] and forecasts with two cooperating paths.

Linear backbone. Per-component linear maps project the window directly to the horizon:

$$\mathbf{Y}_{\text{lin}} = \mathbf{W}_s \mathbf{X}_s + \mathbf{W}_t \mathbf{X}_t \in \mathbb{R}^{H \times C}, \qquad \mathbf{W}_s, \mathbf{W}_t \in \mathbb{R}^{H \times L}, \tag{3}$$

i.e., a DLinear-style map [11] (randomly initialized and trained jointly — a parameterized pathway, not a fitted DLinear), shared across variates.

Selective-scan correction. The pair $[\mathbf{X}_s; \mathbf{X}_t]$ is embedded per time step and processed channel-independently by a stack of Mamba layers [6] (Fig. 2a). Instantiating the selective recurrence of Section 3.2, the per-step discrete parameters are $\bar{\mathbf{A}}_k = \exp(\boldsymbol{\Delta}_k \mathbf{A})$ and $\bar{\mathbf{B}}_k = \boldsymbol{\Delta}_k \mathbf{B}_k$, and the layer runs the recurrence

$$\mathbf{h}_k = \bar{\mathbf{A}}_k \, \mathbf{h}_{k-1} + \bar{\mathbf{B}}_k \, u_k, \qquad y_k = \mathbf{C}_k^{\top} \mathbf{h}_k + D \, u_k, \tag{4}$$

scanned causally over the L steps, independently per feature dimension (diagonal $\mathbf{A} \in \mathbb{R}^N$, scalar input, with $\boldsymbol{\Delta}_k, \mathbf{B}_k, \mathbf{C}_k$ produced from u by learned linear maps — the input-dependent selectivity of [6]). The last state, summed with a whole-window linear embedding $\mathbf{W}_e \tilde{\mathbf{x}}^{(c)}$ (one per variate, as in [7]), forms the branch feature $\mathbf{f}_{\text{time}} \in \mathbb{R}^{C \times d}$. A head ϕ_{time} maps features to a horizon correction:

$$\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}} + \phi_{\text{time}}(\mathbf{f}_{\text{time}}), \qquad \phi_{\text{time}} \equiv \mathbf{0} \text{ at init.} \tag{5}$$

Because ϕ_{time} is zero-initialized, the branch's output at initialization equals its (randomly initialized) DLinear-style backbone; the SSM learns only what the linear map cannot express.

4.2. Frequency Branch: Spectral Filtering and a Dormant Spectral Pathway

The frequency branch applies the real FFT along time, $\mathbf{Z} = \mathrm{rFFT}(\tilde{\mathbf{X}}) \in \mathbb{C}^{C\times F}$, $F = \lfloor L/2 \rfloor + 1$, optionally low-passed by zeroing the top fraction ρ of bins.

Spectral encoder. A learnable complex linear filter $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i \in \mathbb{C}^{F\times F}$ densely mixes bins, $\mathbf{Z}' = \mathbf{Z}\mathbf{W}^\top$ along the frequency axis (a bidirectional Mamba over the bin sequence is an input-dependent alternative). The (real, imaginary) parts are projected to the branch feature $\mathbf{f}_{\mathrm{freq}} \in \mathbb{R}^{C\times d}$ and a head ϕ_{freq} produces the spectral forecast.

FITS-style spectral pathway (optional, per dataset). A complex linear map $\mathbf{U} \in \mathbb{C}^{F'\times F}$ with $F' = \lfloor (L{+}H)/2 \rfloor + 1$, applied along the frequency axis ($\mathbf{Z}\mathbf{U}^\top \in \mathbb{C}^{C\times F'}$), maps the (low-passed) window spectrum to that of a length-($L{+}H$) series, whose inverse transform's last H samples are the branch's linear spectral forecast:

$$\mathbf{Y}_{\mathrm{freq}} = \underbrace{\mathrm{irFFT}_{L+H}(\mathbf{Z}\mathbf{U}^\top)[\,L{:}\,]}_{\text{linear-in-frequency}} + \phi_{\mathrm{freq}}(\mathbf{f}_{\mathrm{freq}}). \tag{6}$$

This uses the frequency-interpolation idea of FITS [12] but is not a fitted FITS model. Where the backbone is enabled (ETTh1, ETTh2, Weather), $\mathbf{U}$ and the head ϕ_{freq} are both zero-initialized, so the branch is silent at init; a zero rather than identity init is used because the length-L and length-($L{+}H$) frequency grids do not align. Elsewhere the backbone is off and ϕ_{freq} is *randomly* initialized, so the branch contributes a small gate-attenuated signal from the first step. The clean RLinear-class start [15, 16] — in which every deep path is zero at init and the model reduces to $g_0 \cdot \mathbf{Y}_{\mathrm{lin}}$ — therefore holds exactly only for the backbone-enabled configurations, and approximately elsewhere, up to a $(1-g_0)$-weighted random term that the near-saturated gate ($g_0 \approx 0.9$) keeps small.

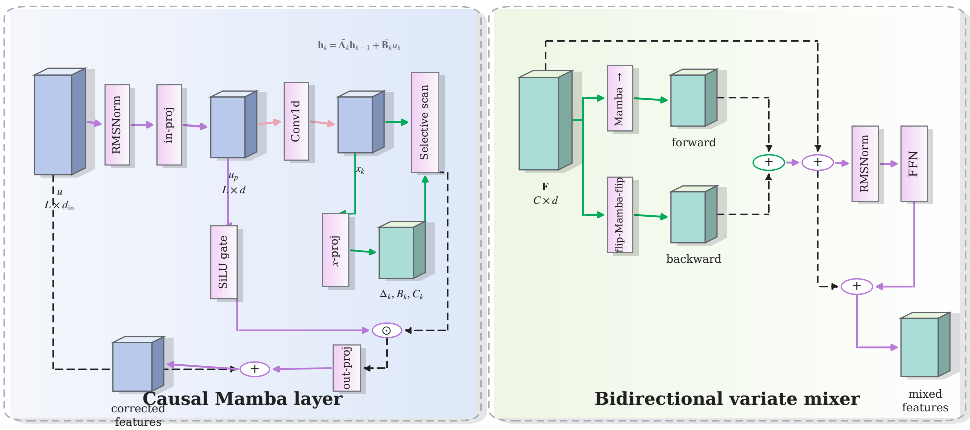

Figure 2: Selective state-space modules used by DD-Mamba. **(a)** The causal time-axis Mamba block normalizes and projects the input, forms a convolutional content stream and a SiLU gate, predicts $(\Delta_k, \mathbf{B}_k, \mathbf{C}_k)$ from the input, and evaluates Eq. (4) with an outer residual connection. **(b)** The variate mixer scans the C channel tokens in both directions, restores the reverse scan to the original order, sums the two streams, and applies residual-normalized position-wise feed-forward refinement. The two panels distinguish temporal causality (ordered time steps) from non-causal cross-channel interaction (channel order has no temporal interpretation).

4.3. Cross-Channel Mixing over Variates

Channel independence is a strong baseline [4, 11] but leaves accuracy on the table when variates are coupled [5, 7, 14]. Inside each branch, before its head, we optionally run M layers of *bidirectional* Mamba over the C per-channel features, followed by a position-wise feed-forward block (Fig. 2b). Because channel order carries no causal direction, each layer sums a forward and a reversed scan in one residual stream. This construction is non-causal, but it is *not* permutation-invariant: an arbitrary reordering of channels can change both scans. We retain each dataset's canonical column order and treat order sensitivity as a limitation rather than attributing set-like symmetry to the mixer.

As a descriptive measure of channel coupling, we compute on the training split

$$\overline{|r|} = \frac{2}{C(C-1)} \sum_{1 \leq i < j \leq C} \left|\mathrm{corr}(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})\right|. \tag{7}$$

It spans 0.31–0.35 on the ETT family and 0.50–0.91 on Electricity, Traffic, and Solar (Table 1). Raw PCC does not represent lagged, nonlinear, or conditional dependence and can be inflated by common trends; it is therefore used only to organize the analysis. The mixer remains a per-dataset switch whose value is evaluated empirically in Section 5.3.

4.4. Forecast-Level Convex Fusion

Each branch emits a forecast and a feature. The model output is a gated convex combination

$$\mathbf{Y} = g \odot \mathbf{Y}_{\text{time}} + (1-g) \odot \mathbf{Y}_{\text{freq}}, \qquad g = \sigma\big(\psi([\mathbf{f}_{\text{time}}; \mathbf{f}_{\text{freq}}])\big) \in (0,1)^{C\times 1}, \tag{8}$$

with ψ applied row-wise (per variate), followed by RevIN de-normalization. Since $\mathbf{Y}$ lies elementwise between the branch forecasts, both domains retain an explicit prediction path, and the gate value records the convex mixing weight. It is neither an identifiable component contribution nor a causal attribution, because the branches are jointly trained and can compensate for one another. The constraint is structural, not behavioral: g is a sigmoid and can saturate near 0 or 1, so one branch may contribute very little — by design, as the ablations show the useful domain is dataset-dependent. ψ is zero-initialized with its bias set so that training starts gate-open toward the time branch ($g_0 \approx 0.9$; exact value in Section 5.1): the time branch is linear at initialization while the frequency branch is silent, and an even initial mix would halve the initial linear forecast. This is an initialization convention; empirical gate distributions are not available in the present study.

4.5. Training

All branches, gates, and heads are optimized jointly by minimizing the MSE between the forecast and the target. Optimizer, schedule, stopping rule, and dataset-specific settings are reported with the experimental protocol in Section 5.1.3.

5. Experiments

5.1. Setup

5.1.1. Datasets and Protocol

We evaluate on nine standard benchmarks (Table 1) under the unified protocol of [5, 7]: look-back 96, horizons 96/192/336/720. ETT uses the canonical 12/4/4-month border split, the others chronological 0.7/0.1/0.2 splits; variates are standardized by training-set statistics; metrics are MSE and MAE.

5.1.2. Baselines

We compare linear, attention-based, and state-space forecasters. The primary comparison uses one evaluation pipeline (Table 4): DLinear [11],

Table 1: Benchmark datasets, their cross-channel coupling, and where DD-Mamba places its capacity. $\overline{|r|}$: mean absolute pairwise Pearson correlation between channels on the training split (— = raw file unavailable in our build environment).

Dataset	Variates	Steps	Granularity	$\overline{\lvert r\rvert}$	Variate mixer	Params
ETTh1/ETTh2	7	17,420	1 hour	0.31/0.35	off	0.25M
ETTm1/ETTm2	7	69,680	15 min	0.31/0.35	off	0.24–0.36M
Weather	21	52,696	10 min	—	M=2, d=256	5.8M
Solar-Energy	137	52,560	10 min	0.91	M=1, d=128	0.96M
Electricity	321	26,304	1 hour	0.50	M=2, d=512	19.7M
Traffic	862	17,544	1 hour	0.58	M=2, d=512	19.7M
Exchange	8	7,588	1 day	0.48	off	0.09M

NLinear [11], RLinear [15], PatchTST [4], iTransformer [5], S-Mamba [7], and ms-Mamba [8] are re-implemented in our codebase and trained through the same data pipeline, splits, schedule, and evaluation as DD-Mamba, over five matched run indices, on six datasets (ETTh1, ETTh2, ETTm1, ETTm2, Weather, Electricity). These controls reduce split and evaluation confounds but do not equalize model-specific tuning budgets.

Four further baselines from the standard comparison set — Crossformer [14], TiDE [19], FEDformer [3] and Autoformer [2] — are implemented in the same harness (`src/models/baselines.py`) and included in the default sweep, but their runs are not yet complete. They therefore appear only in Table 6, at their *published* values, and not in the same-pipeline Table 4; we do not transcribe published figures into the controlled comparison.

Published results are not mixed with this ranking because their preprocessing, optimization, and seeds differ. Crossformer [14] is a particularly relevant spatial–temporal baseline, but our adapter omits its hierarchical encoder–decoder merging and is therefore not presented as a faithful Crossformer result. The S-Mamba and ms-Mamba entries are likewise repo-native reproductions rather than outputs from the authors' code; our ms-Mamba adapter realizes the multi-scale idea by *temporally subsampling* the lookback at rates $\{1, 2, 4\}$ and summing per-scale Mamba encoders, rather than the published design's parallel Mamba blocks with distinct per-scale state-space sampling-rate (Δ) parameters, so it is a multi-scale proxy. The accuracy claim is consequently restricted to the seven implemented baselines and six datasets of Table 4; no superiority claim is made against Crossformer, TF4TF, or the three datasets without in-pipeline comparisons.

5.1.3. Implementation and Reproducibility.

All experiments use PyTorch 2.11 on a single NVIDIA RTX 5090. Optimization is AdamW for at most 10 epochs, with the learning rate halved after each epoch, gradient clipping at 5.0, and best-checkpoint early stopping on validation loss. The decomposition kernel is 25; Mamba blocks use expansion 2 and convolution width 4. The search space, selected configurations, and remaining implementation details are reported in Table 2 and the reproducibility package. The schedule is held fixed across runs; Section 6.3 reports the sweep that tested it against a longer warm-up-plus-cosine budget. DD-Mamba's per-horizon results (Table 3) are means over *three seeds* (2024–2026); the maximum per-entry std across the 36 dataset–horizon entries is 0.005 MSE, typically 0.001–0.003. The unified comparison (Table 4) uses *five* matched run indices (2024–2028) for every model, and DD-Mamba's five-seed averages reproduce its three-seed numbers to within 0.002 MSE. Table 2 records final empirical configurations, not a transferable selection rule:

no preregistered search space or equalized per-model tuning budget was used.

5.1.4. Statistical Methodology

We compare at two levels. *Within our framework* (ablations), every variant shares seeds with the reference, so we form the *paired* per-seed difference and report its mean effect, a two-sided t-based p-value, and a 95% CI (n=3, t_{crit}=4.303). Because we run many such tests, we control the false discovery rate with the Benjamini–Hochberg (BH) procedure *within each family* — the 29 component-placement comparisons, the 54 branch-removal comparisons, and the 11 PEMS04 comparisons — and report the resulting q-values (`scripts/compute_stats_correction.py`). An effect is called statistically supported only if it survives BH at q<0.05 and its configuration was not selected using the same test set; all other effects are reported as exploratory. We flag n=3 as a substantial limitation of power, and report effect sizes with CIs throughout so that magnitude can be judged independently of the significance verdict. *Unified in-pipeline comparison* (Table 4): we test each DD-Mamba-vs-baseline difference on average-over-horizons MSE using the five matched run indices and a paired two-sided t-test (t_{crit}=2.776), followed by BH correction across the comparison family (`scripts/analyze_unified_baselines.py`). The paired interpretation relies on matching run index and data order.

Table 2: Per-dataset hyperparameters. d: model width; time enc.: time-branch encoder (Mamba layers in parentheses); M: variate-mixer layers; N: SSM state size; ρ: spectral low-pass fraction (Section 4.2); spec. map: the *optional* FITS-style pathway of Eq. (6) — when "no", the frequency branch remains fully active (rFFT, spectral encoder, forecast head) but carries no explicit linear pathway; IN: instance normalization (RevIN); do.: dropout; B: batch size; wd: weight decay. †Solar adopts IN = no after inspection of a test-set ablation (Section 5.3); all results under this setting are labeled exploratory and excluded from confirmatory claims. The widths listed here are the configurations that produced the reported accuracy, and the repository's released configs match them. The variate mixer is instantiated independently in both branches (`mixer_placement=both`) except on Electricity and Traffic, where weight-tying it across branches was measured accuracy-neutral and is used to halve the mixer cost (Section 6). Parameter-reduced variants of these models, and the accuracy they cost, are reported separately in Table 12; they are not the configurations behind the results tables. §Electricity's reported numbers were produced with the spectral pathway disabled and therefore with a randomly initialized frequency head; Section 6 argues this accounts for part of its short-horizon deficit. Enabling the pathway was tested and made the dataset average worse (0.169 → 0.175, almost entirely at H=720), so it remains disabled.

Dataset	d	time enc.	M	N	ρ	spec. map	IN	lr	B	do./wd
ETTh1	128	Mamba (1)	0	16	0.4	yes	yes	$5{\cdot}10^{-4}$	32	$0.2/10^{-4}$
ETTh2	128	Mamba (1)	0	16	0.3	yes	yes	$5{\cdot}10^{-4}$	32	$0.3/5{\cdot}10^{-4}$
ETTm1	128	Mamba (2)	0	16	0	no	yes	10^{-4}	32	$0.1/10^{-4}$
ETTm2	128	Mamba (1)	0	16	0.2	no	yes	10^{-4}	32	$0.2/2{\cdot}10^{-4}$
Weather	256	Mamba (2)	2	16	0	yes	yes	$2{\cdot}10^{-4}$	32	$0.1/10^{-4}$
Solar	128	Mamba (2)	1	16	0	no	no†	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Electricity	512	MLP	2	32	0	no§	yes	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Traffic	512	MLP	2	32	0	no	yes	10^{-3}	16	$0.1/10^{-4}$
Exchange	64	Mamba (1)	0	8	0.5	no	yes	10^{-4}	32	$0.3/5{\cdot}10^{-4}$

Table 3: DD-Mamba per-horizon results, look-back L=96, from the most recent run of the released configurations. Solar uses a test-informed normalization setting and is reported only as an exploratory diagnostic result. Values are point estimates from this run; the seed-level dispersion quoted elsewhere in the paper comes from the three-seed study of the same configurations, whose per-horizon means agree with these to within ±0.003 MSE on every dataset. Electricity and Weather were re-run at the capacity Table 2 lists after the reduction study; Table 12 gives the separately measured cost of their parameter-reduced variants.

Dataset		H=96	H=192	H=336	H=720	Avg
ETTh1	MSE	0.381	0.429	0.472	0.472	0.439
	MAE	0.397	0.423	0.446	0.468	0.434
ETTh2	MSE	0.292	0.368	0.415	0.423	0.374
	MAE	0.345	0.392	0.427	0.442	0.402
ETTm1	MSE	0.321	0.366	0.396	0.461	0.386
	MAE	0.359	0.385	0.405	0.443	0.398
ETTm2	MSE	0.177	0.241	0.302	0.398	0.280
	MAE	0.260	0.302	0.341	0.397	0.325
Weather	MSE	0.162	0.212	0.267	0.349	0.247
	MAE	0.207	0.254	0.293	0.347	0.275
Electricity	MSE	0.141	0.159	0.174	0.200	0.169
	MAE	0.236	0.253	0.270	0.296	0.264
Traffic	MSE	0.401	0.423	0.441	0.481	0.437
	MAE	0.268	0.276	0.284	0.303	0.283
Exchange	MSE	0.086	0.179	0.329	0.858	0.363
	MAE	0.206	0.301	0.415	0.699	0.405
Solar	MSE	0.181	0.189	0.207	0.206	0.196
	MAE	0.241	0.256	0.269	0.269	0.259

Table 4: *Unified in-pipeline* comparison, **Parts II and III** of the main results (average MSE/MAE over $H \in \{96, 192, 336, 720\}$, look-back L=96; five matched run indices; lower is better). Part I — the traffic-flow benchmarks — is Table 5, which is reported separately because its competitor figures are published rather than re-run. Every baseline here is re-implemented and evaluated with the same splits and protocol as DD-Mamba; model-specific tuning budgets were not equalized. Following the presentation of S-Mamba [7] and ms-Mamba [8], the **best** result per dataset and metric is bold and the second best underlined. * DD-Mamba beats the best competing baseline at Benjamini–Hochberg q<0.05 (paired t-test, n=5; `analyze_unified_baselines.py`); those papers report point estimates without significance testing, so the starred cells are the subset of wins this protocol can actually support. "—" marks a dataset on which the baseline sweep has not been run; those rows give DD-Mamba's own result and support no comparison. † Solar uses a test-informed normalization setting (Section 6.3) and is exploratory. DD-Mamba has the lowest MSE on five of the six datasets with baselines and the corrected test supports the margin on four; Weather is a statistical tie (q=0.23) and PatchTST is better on ETTm1.

Models	**DD-Mamba**		S-Mamba		ms-Mamba		iTransformer		PatchTST		RLinear		NLinear		DLinear	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Part II — ETT benchmarks (7 variates).																
ETTh1	**0.439***	**0.434**	0.463	0.458	0.459	0.451	0.444	0.442	0.456	0.449	0.457	0.443	0.464	0.450	0.462	0.456
ETTh2	**0.376***	**0.401**	0.394	0.410	0.409	0.420	0.383	0.407	0.385	0.408	0.391	0.411	0.390	0.412	0.545	0.509
ETTm1	0.387	0.398	0.397	0.406	0.393	0.403	0.407	0.409	**0.381**	**0.396**	0.424	0.413	0.429	0.419	0.410	0.412
ETTm2	**0.280***	**0.325**	0.297	0.335	0.288	0.333	0.292	0.335	0.284	0.331	0.293	0.336	0.294	0.337	0.359	0.407
Part III — remaining benchmarks (8–321 variates).																
Weather	**0.248**	0.276	0.251	0.278	0.257	0.281	0.265	0.285	0.249	**0.275**	0.273	0.292	0.279	0.296	0.267	0.318
Electricity	**0.168***	**0.263**	0.183	0.275	0.187	0.277	0.195	0.282	0.189	0.276	0.217	0.295	0.220	0.299	0.211	0.299
Exchange	0.363	0.405	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Solar†	0.196	0.259	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 5: **Part I of the main results: the traffic-flow benchmarks** (MSE, look-back L=96). The PEMS blocks use the short-horizon grid $T \in \{12, 24, 48, 96\}$ standard for those datasets and compare against the *published* numbers of S-Mamba [7] and ms-Mamba [8]; the Traffic block uses the long-horizon grid $H \in \{96, 192, 336, 720\}$ shared with Parts II and III (Table 4), where no in-pipeline baseline sweep has been run, so only DD-Mamba's own result is available. **Best** bold, <u>second best</u> underlined. **These are not same-pipeline reruns.** Unlike Table 4, the competitors' figures are transcribed from their papers and were produced by their own code, preprocessing and seeds, so this table is a positioning reference rather than a controlled comparison, and no significance test is applicable to it. PEMS is a confirmatory family for us: these datasets were not used to develop any component of DD-Mamba.

Dataset	Model	T=12	T=24	T=48	T=96	Avg
PEMS03	DD-Mamba	<u>0.065</u>	**0.085**	**0.128**	**0.187**	**0.116**
	S-Mamba	**0.065**	<u>0.087</u>	<u>0.133</u>	<u>0.201</u>	<u>0.122</u>
	ms-Mamba	0.066	<u>0.087</u>	<u>0.133</u>	<u>0.201</u>	0.122
PEMS04	DD-Mamba	0.075	0.093	0.123	0.171	0.116
	S-Mamba	<u>0.076</u>	<u>0.084</u>	<u>0.115</u>	<u>0.137</u>	<u>0.103</u>
	ms-Mamba	**0.072**	**0.083**	**0.099**	**0.121**	**0.094**
PEMS07	DD-Mamba	0.059	0.079	0.105	0.139	0.095
	S-Mamba	<u>0.063</u>	<u>0.081</u>	<u>0.093</u>	<u>0.117</u>	<u>0.089</u>
	ms-Mamba	**0.060**	**0.075**	**0.091**	**0.109**	**0.084**
PEMS08	DD-Mamba	**0.074**	**0.099**	**0.146**	**0.234**	**0.138**
	S-Mamba	<u>0.076</u>	0.104	0.167	0.245	0.148
	ms-Mamba	<u>0.073</u>	<u>0.098</u>	<u>0.154</u>	<u>0.236</u>	<u>0.140</u>
Traffic (862 variates), long-horizon grid:						
		H=96	H=192	H=336	H=720	Avg
Traffic	DD-Mamba	0.401	0.423	0.441	0.481	0.437
	S-Mamba	—	—	—	—	—
	ms-Mamba	—	—	—	—	—

Table 6: Positioning against the *published* benchmark compilation of iTransformer [5], which S-Mamba [7] and ms-Mamba [8] also report against. Averages over $H \in \{96, 192, 336, 720\}$ ($T \in \{12, 24, 48, 96\}$ for PEMS), look-back L=96; ETT and PEMS are averaged over their four datasets, matching the granularity at which the source reports them. This table spans all three parts of the comparison and is the only place the full standard baseline set appears. **It is not a controlled comparison.** Every competitor figure was produced by a different codebase, preprocessing and seed set, so the numbers are a positioning reference and carry no significance test; the same-pipeline evidence is Table 4. The size of the confound is measurable: re-running four of these baselines in our pipeline (Section 5.1) moves them by up to 0.017 MSE in either direction, which exceeds several of the margins below. **Best** bold, <u>second best</u> underlined — under that caveat. S-Mamba's own column is omitted because its paper could not be retrieved in our build environment; its published PEMS figures are in Table 5 and its same-pipeline rerun in Table 4. † Solar uses a test-informed normalization setting and is exploratory.

Models	**DD-Mamba**		iTransformer		RLinear		PatchTST		Crossformer		TiDE		DLinear		FEDformer		Autoformer	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Traffic	<u>0.437</u>	<u>0.283</u>	**0.428**	**0.282**	0.626	0.378	0.481	0.304	0.550	0.304	0.760	0.473	0.625	0.383	0.610	0.376	0.628	0.379
PEMS (Avg)	**0.116**	**0.217**	<u>0.119</u>	<u>0.218</u>	0.514	0.482	0.217	0.305	0.220	0.304	0.375	0.440	0.320	0.394	0.224	0.327	0.614	0.575
ETT (Avg)	**0.370**	**0.390**	0.383	0.399	<u>0.380</u>	<u>0.392</u>	0.381	0.397	0.685	0.578	0.482	0.470	0.442	0.444	0.408	0.428	0.465	0.459
Weather	**0.247**	**0.275**	<u>0.258</u>	<u>0.278</u>	0.272	0.291	0.259	0.281	0.259	0.315	0.271	0.320	0.265	0.317	0.309	0.360	0.338	0.382
Electricity	**0.169**	**0.264**	<u>0.178</u>	<u>0.270</u>	0.219	0.298	0.205	0.290	0.244	0.334	0.251	0.344	0.212	0.300	0.214	0.327	0.227	0.338
Exchange	0.363	0.405	<u>0.360</u>	**0.403**	0.378	0.417	0.367	<u>0.404</u>	0.940	0.707	0.370	0.413	**0.354**	0.414	0.519	0.429	0.613	0.539
Solar†	**0.196**	**0.259**	<u>0.233</u>	<u>0.262</u>	0.369	0.356	0.270	0.307	0.641	0.639	0.347	0.417	0.330	0.401	0.291	0.381	0.885	0.711

We report the main comparison in three parts, grouped by the kind of data each covers: **Part I**, the traffic-flow benchmarks Traffic and the four PEMS sets (Table 5); **Part II**, the four ETT benchmarks; and **Part III**, the remaining datasets — Weather, Electricity, Exchange and Solar (both in Table 4). The split is not only presentational: Part I is the only group whose competitor numbers are *published* rather than re-run in our pipeline, and it uses the short-horizon grid conventional for PEMS, so its evidential status differs from Parts II and III and it is kept in a separate table. Table 6 then places DD-Mamba against the full standard baseline set at the granularity the published compilation reports, spanning all three parts; the paragraph on cross-pipeline comparability below explains why that table cannot carry a superiority claim.

Part I: traffic-flow benchmarks.. On the four PEMS datasets (Table 5) DD-Mamba is ahead of both published Mamba forecasters on PEMS03 (0.116 vs. 0.122/0.122) and PEMS08 (0.138 vs. 0.148/0.140), and behind both on PEMS04 and PEMS07, where ms-Mamba's multi-scale sampling rates are strongest. Because the competing figures come from their own pipelines, we read this as *positioning* rather than as evidence of superiority: the split is two datasets each way, and the protocol difference is large enough that a 0.005-scale margin should not be over-interpreted. Traffic carries no in-pipeline baseline sweep, so its row records DD-Mamba's own accuracy (0.437 average MSE) and supports no comparison.

Parts II and III: ETT and the remaining benchmarks.. Table 4 is the paper's primary accuracy evidence because it uses one data and evaluation protocol. Under this five-run protocol DD-Mamba has the lowest average MSE on five of the six re-run datasets, and the paired, BH-corrected test supports the margin on **four**: Electricity (0.168 vs. S-Mamba's 0.183; $q \ll 0.01$), ETTh2 (0.376 vs. iTransformer's 0.383; q=0.001), ETTm2 (0.280 vs. PatchTST's 0.284; q=0.015), and ETTh1 (0.439 vs. iTransformer's 0.444; q=0.021). Weather is a statistical *tie* — DD-Mamba is numerically best (0.248 vs. PatchTST's 0.249) but the difference does not survive correction (q=0.23). On ETTm1, PatchTST is lower (0.381 vs. 0.387; q<0.001), consistent with the branch matrix, where ETTm1 is carried by the time branch and its patch-attention competitor is strong. DD-Mamba's size is dataset-dependent: the low-channel ETT and Exchange configurations use only $\approx$ 0.1–0.25M parameters, an order of magnitude fewer than the variate-attention baselines, whereas on high-channel data the variate mixer dominates the count (Table 13). These results do not establish a ranking on Solar, Traffic, or Exchange, where comparable in-pipeline baselines are unavailable.

How far the published comparison can be trusted.. Against the published compilation (Table 6) DD-Mamba has the lowest average error on five of the seven dataset groups, is second on Traffic and third on Exchange. We do *not* read this as evidence of superiority, because we can measure the confound directly: four of those baselines are also re-implemented here, so their published figure and their same-pipeline figure can be compared. On ETT and Weather the two agree closely — DLinear 0.444 vs. 0.442, iTransformer 0.382 vs. 0.383 on ETT — which is a useful check that our pipeline reproduces the standard protocol. On Electricity they do not: our iTransformer scores 0.195 against a published 0.178, and our PatchTST 0.189 against a published 0.205. The pipeline term is therefore worth up to 0.017 MSE and runs in *both* directions, while DD-Mamba's margin over published iTransformer on the same dataset is 0.009. The margin is inside the confound. Table 6 accordingly establishes that DD-Mamba is competitive with the standard baseline set, and nothing stronger; the claims this paper actually defends rest on Table 4, where the pipeline is held fixed. A notable pattern in the reproduction gap is that the linear models transfer almost exactly while the deep ones move most, consistent with the published deep baselines carrying per-dataset tuning that a single shared schedule does not reproduce.

5.3. Component Ablations

Our ablation harness flips exactly one switch of a dataset's full configuration, holding split, schedule, seeds, and every other hyperparameter fixed. The design follows S-Mamba [7], which groups variants into a *reference* row, a *Replace* block (component swapped for an alternative of the same role) and a *w/o* block (component removed), and ms-Mamba [8], which additionally sweeps a design *choice* rather than only deleting parts. Table 7 lays out the resulting grid over the three blocks of DD-Mamba: the cross-variate mixer (S-Mamba's *variate correlation* block), the time-axis encoder (their *temporal dependency* block), and the frequency branch, which has no S-Mamba counterpart. We implement S-Mamba's two mixer alternatives, multi-head attention and a unidirectional scan (`mixer_kind`), so that comparison runs inside our pipeline rather than being inherited across papers; and because a dataset already fixes several switches (PEMS ships an MLP time encoder and a weight-tied mixer, ETT ships Mamba and untied), the grid is resolved

Table 7: Ablation design, following S-Mamba's grouping into a reference row, *Replace* rows (component swapped for an alternative of the same role) and *w/o* rows (component removed). VC: cross-variate mixer; TD: time-axis encoder; FD: frequency branch. "base" is whatever the dataset's released configuration specifies, so the grid resolves to 12 runs on the high-channel datasets and 7 where no mixer is configured.

Design	Variant	VC	TD	FD
reference	full	bi-Mamba	base	linear filter
Replace	uni-directional mixer	uni-Mamba	base	linear filter
	attention mixer	Attention	base	linear filter
	mixer placement	tied $\leftrightarrow$ untied	base	linear filter
	mixer in time branch only	time only	base	linear filter
	time encoder	bi-Mamba	Mamba $\leftrightarrow$ MLP	linear filter
	spectral encoder	bi-Mamba	base	bi-Mamba
	fusion rule	bi-Mamba	base	linear filter (averaged)
w/o	no cross-variate mixing	w/o	base	linear filter
	no frequency branch	bi-Mamba	base	w/o
	no time branch	bi-Mamba	w/o	linear filter
	no instance normalization	bi-Mamba	base	linear filter

per configuration, so no row silently duplicates the reference and inflates the correction family.

One methodological difference matters: S-Mamba and ms-Mamba report single-run deltas, whereas we use multiple seeds, paired CIs and Benjamini–Hochberg across the family — with a dozen variants, about one would clear an uncorrected threshold by chance. We report the harness's complete output on three datasets under their *final released configurations*, three seeds each at H=96: ETTh1 and Solar (Table 8) and Weather (Table 10). Two consistency checks pass: the Solar "full" row reproduces the main-table Solar H=96 entry (0.180$\pm$.004), and its w/o-instance-norm variant coincides with

Table 8: Component ablations on ETTh1 and Solar-Energy (H=96, three seeds, final configurations): paired per-seed ΔMSE to the full model with t-based 95% CIs; * CI excludes zero. The spectral map is enabled on ETTh1 (hence *removed*) and disabled on Solar (hence *added*); the variate mixer is off on ETTh1 (*added*) and on for Solar (*removed*).

Variant	ETTh1: Δ [95% CI]	Solar: Δ [95% CI]
full (reference MSE)	0.3802$\pm$.001	0.1802$\pm$.004
time branch only (freq removed)	+0.0034 [−0.009, +0.016]	+0.0114 [+0.001, +0.021]*
freq branch only (time removed)	−0.0007 [−0.004, +0.003]	+0.0051 [−0.013, +0.023]
sum fusion	−0.0026 [−0.007, +0.002]	+0.0066 [−0.005, +0.018]
w/o instance norm	+0.0061 [−0.001, +0.013]	—
w/o linear backbone	−0.0031 [−0.009, +0.003]	+0.0081 [−0.019, +0.035]
random init (vs. zero-init)	−0.0012 [−0.005, +0.002]	−0.0046 [−0.022, +0.013]
spectral map removed / added	+0.0042 [−0.018, +0.026]	+0.0088 [−0.002, +0.019]
variate mixer added / removed	+0.0077 [+0.004, +0.012]*	+0.0119 [−0.010, +0.034]
Mamba→MLP over time	−0.0022 [−0.007, +0.003]	+0.0089 [−0.002, +0.020]
freq Mamba encoder	−0.0024 [−0.011, +0.006]	−0.0117 [−0.044, +0.020]

the full (RevIN-off) configuration, so that cell is omitted.

Component-placement effects are directional but exploratory.. A two-sided pattern spans datasets at the raw level: adding the variate mixer on ETTh1 *hurts* (by +0.0077 MSE, raw p=0.014) while removing it on Weather also hurts (by +0.0104, raw p=0.022); normalization shows the same shape (removing it hurts Weather +0.0161, raw p=0.023, while disabling it on Solar improved the re-measured main results from 0.228 to 0.199). Across the 29 component-placement comparisons, however, *none* survives Benjamini–Hochberg correction (the five raw-significant effects have $q \geq 0.11$): at three seeds these are *exploratory, directional* findings. Zero initialization is a *null* result on both datasets (−0.0012 ETTh1, −0.0046 Solar; CIs include zero), so we claim no accuracy benefit for it, only the interpretable linear start

(Section 4.1). The DLinear backbone and gate-versus-average are within CIs everywhere; on Weather three components have unadjusted CIs excluding zero (normalization, the variate mixer, and the temporal Mamba). The recurring pattern — component value is dataset-dependent in *both* directions — is the argument for per-dataset switches over universal defaults.

Branch matrix: where each domain matters.. Because a single Solar cell cannot carry a dual-domain conclusion, we ran the full branch-ablation matrix — full / time-only / freq-only under paired seeds on all nine datasets at H=96 and across all four horizons on the six ETT/Weather/Solar datasets, 27 cells in total (Table 9). Every full-model mean reproduces the corresponding main-table entry, and the Solar H=96 cell replicates the earlier ablation (+0.0126 vs. +0.0114, overlapping CIs). The 27 cells yield 54 tests; we report raw significance (* in Table 9) and BH-corrected q-values, visualized in Fig. 3.

Frequency branch. Removing it is raw-significant in 5 cells, concentrated at the shortest horizon: Solar (+0.0126), ETTm1 (+0.0073), Electricity (+0.0019), and Exchange (+0.0015) at H=96, plus ETTm2 at H=336. After correction only **Solar at** H=96 survives (q=0.048); the other four fall to $q \geq 0.10$. The frequency-side result is therefore narrow and, because Solar's normalization setting was selected with test feedback, remains *exploratory.*

Time branch. Removing it is raw-significant in 9 cells and **8 survive BH** (q<0.05): ETTm1 at *every* horizon (+0.017–+0.021, the most persistent effect we measure), ETTm2 at H=96 and H=336, and Electricity and Exchange at H=96. On ETTh1, ETTh2, and Weather no branch effect is detected at any horizon; this is absence of detected evidence, not evidence

Table 9: Branch-ablation matrix (27 cells): paired per-seed ΔMSE versus the full model when one branch is removed (three seeds; * marks *raw* t-based 95% CI excluding zero; per-cell CIs ship with the results). After Benjamini–Hochberg correction over the 54 tests, 9 survive at $q<0.05$ — eight on the time side (ETTm1 all horizons; ETTm2@96/336; Electricity@96; Exchange@96) and, on the frequency side, *only* Solar@96 (q=0.048); the remaining starred cells are exploratory. The Solar result is also exploratory because its normalization setting was selected with test feedback. Electricity, Traffic, and Exchange were run at H=96 only (—). Full-model means match Table 3.

	freq removed: ΔMSE at H				time removed: ΔMSE at H			
Dataset	96	192	336	720	96	192	336	720
ETTh1	+.0034	+.0005	−.0047	+.0096	−.0007	−.0001	−.0003	+.0176
ETTh2	+.0006	+.0078	+.0054	+.0093	−.0031	−.0017	−.0017	+.0026
ETTm1	+.0073*	+.0002	+.0019	−.0005	+.0196*	+.0168*	+.0203*	+.0210*
ETTm2	+.0020	+.0014	+.0030*	+.0001	+.0060*	+.0060	+.0059*	+.0059
Weather	+.0008	+.0004	−.0002	−.0009	+.0006	+.0011	+.0021	+.0027
Solar	+.0126*	−.0038	+.0006	−.0032	+.0052	+.0075	+.0040*	+.0027
Electricity	+.0019*	—	—	—	+.0017*	—	—	—
Traffic	−.0015	—	—	—	+.0013	—	—	—
Exchange	+.0015*	—	—	—	+.0058*	—	—	—

that the branches are interchangeable.

The matrix thus shows repeated sensitivity to the temporal branch and only a test-informed frequency-side effect, not a consistent dual-domain advantage. A branch with a small fitted gate weight remains computationally active, so it should not be described as cost-free redundancy.

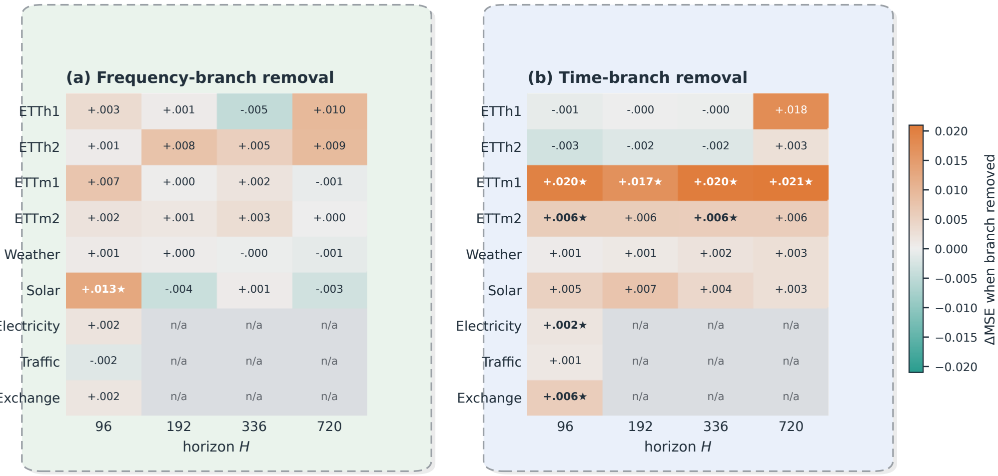

Figure 3: Branch-removal effect (ΔMSE) across datasets and horizons, visualizing Table 9: **(a)** removing the frequency branch and **(b)** removing the time branch. Warm cells mark configurations where removal *hurts* (the branch is valuable), cool cells where removal *helps* (the branch is redundant), and gray where there is no effect; a ★ marks effects surviving Benjamini–Hochberg correction ($q < 0.05$). Only Solar@96 survives on the frequency side, whereas the time branch is confirmed across the ETTm datasets and at H=96 on Electricity and Exchange. "n/a": not run.

PEMS04: the first dataset where components separate.. On the nine original benchmarks no component effect survived correction. On PEMS04 (Table 11) six of eleven do, and the pattern is coherent. Cross-variate mixing dominates: removing it costs +0.0232 MSE, a 31% degradation and by far the largest effect we have measured. Its *form* also matters — replacing the bidirectional Mamba with attention costs +0.0062 and with a unidirectional scan +0.0010, so on this dataset our pipeline independently reproduces S-Mamba's two design claims [7]: a selective state space beats attention over the variate

Table 10: Component ablation on Weather (H=96; MSE mean ± std over three seeds). Δ is the *paired* per-seed difference to the full model with a t-based 95% confidence interval (n=3); * CI excludes zero.

Variant	MSE	paired Δ [95% CI]	Component isolated
full	0.1620±.0003	—	reference
w/o instance norm	0.1781±.0035	+0.0161 [+0.0053, +0.0269]*	RevIN
w/o channel mixer	0.1724±.0021	+0.0104 [+0.0037, +0.0172]*	cross-variate Mamba mixing
Mamba→MLP over time	0.1647±.0005	+0.0027 [+0.0020, +0.0034]*	selective scan (time)
w/o linear backbone	0.1631±.0006	+0.0011 [−0.0014, +0.0036]	DLinear anchor (time)
time branch only	0.1628±.0011	+0.0007 [−0.0020, +0.0035]	frequency branch removed
w/o spectral map	0.1627±.0016	+0.0007 [−0.0041, +0.0055]	FITS-style pathway
freq branch only	0.1626±.0014	+0.0006 [−0.0040, +0.0052]	time branch removed
freq Mamba encoder	0.1626±.0016	+0.0006 [−0.0044, +0.0055]	spectral encoder type
sum fusion	0.1622±.0007	+0.0001 [−0.0020, +0.0023]	learned gate → average

axis, and the scan should be bidirectional because channel order carries no temporal meaning. Mixing in both branches beats mixing only in the time branch (+0.0013), while tying the two mixers is neutral (−0.0002), matching the efficiency result of Section 6.

The frequency branch does not separate here: removing it costs +0.0006 with a CI spanning zero, and swapping its filter for a bidirectional Mamba changes nothing detectable, whereas removing the *time* branch does hurt (+0.0012†). The asymmetry matches the branch matrix on the original benchmarks.

Two results run against the shipped configuration. Replacing the MLP time encoder with Mamba *improves* PEMS04 (−0.0010, CI excludes zero but q=0.052), and disabling instance normalization improves it substantially (−0.0038†, a 5% MSE reduction). We adopt neither: both are measured on the test split, so acting on them would repeat the test-informed selection

Table 11: S-Mamba-style component ablation on PEMS04 (T=12, three seeds), following the grid of Table 7. ΔMSE is the paired per-seed change against the full model (positive = worse), with t-based 95% CIs. * CI excludes zero; † survives Benjamini–Hochberg across the eleven-variant family at q<0.05. Reference: MSE 0.0754, MAE 0.1794.

Design	Variant	MSE	MAE	ΔMSE	95% CI
Replace	uni-Mamba mixer	0.0763	0.1799	+0.0010†	[+0.0005, +0.0015]
	Attention mixer	0.0816	0.1842	+0.0062†	[+0.0047, +0.0078]
	untie the two mixers	0.0752	0.1795	−0.0002	[−0.0018, +0.0014]
	mixer in time branch only	0.0767	0.1804	+0.0013†	[+0.0007, +0.0020]
	time encoder: MLP → Mamba	0.0744	0.1778	−0.0010*	[−0.0018, −0.0002]
	spectral filter → bi-Mamba	0.0759	0.1801	+0.0005	[−0.0016, +0.0026]
	plain average instead of gate	0.0757	0.1804	+0.0003*	[+0.0000, +0.0006]
w/o	no variate mixing	0.0985	0.2058	+0.0232†	[+0.0228, +0.0235]
	no frequency branch	0.0759	0.1805	+0.0006	[−0.0008, +0.0020]
	no time branch	0.0766	0.1812	+0.0012†	[+0.0004, +0.0020]
	no instance norm	0.0715	0.1755	−0.0038†	[−0.0052, −0.0024]

that makes the Solar normalization setting exploratory. They are recorded as validation-only experiments for future work.

Negative results that shaped the design.. Deepening the traffic mixer from two to four layers (19.7 → 37.9M) gave no gain, so the release keeps two; enabling the mixer on ETT (8.4K training windows) tripled parameters and was associated with premature early stopping and worse held-out error, so the released ETT configurations disable it.

5.4. Analysis

Where branch removals matter.. The matrix (Table 9) shows that the temporal path has the more repeatable effect: eight removals survive correction, including every ETTm1 horizon. The sole corrected frequency-side cell is Solar at H=96, where the 10-minute daily cycle spans $144 > L = 96$ samples. This periodicity offers a plausible hypothesis for the short-horizon effect, but the mechanism is not identified by the ablation and the result remains exploratory because the Solar configuration used test feedback. On ETTh1, ETTh2, Weather, and Traffic, the study does not detect a branch-removal effect under the tested settings. Such null results should not be interpreted as equivalence, especially at three seeds.

Cross-channel correlation as a descriptive covariate.. The mixer-off ETT family has $\overline{|r|}$ = 0.31–0.35, whereas Electricity, Traffic, and Solar, whose released configurations include a mixer, have 0.50–0.91. This alignment motivates a correlation-to-capacity hypothesis but does not validate one: Weather lacks a measured PCC, Exchange has $\overline{|r|} = 0.48$ with the mixer disabled, the switch settings were not generated by a preregistered threshold, and raw PCC can reflect common trends rather than exploitable conditional dependence. We therefore report PCC as a dataset descriptor; a predictive selector would require lag-aware and nonlinear dependence measures, validation-only threshold selection, and evaluation on held-out datasets.

The error profile is horizon-dependent.. On Electricity the margin against the strongest published competitor varies monotonically with the horizon: relative to the S-Mamba average, DD-Mamba is 1.4% worse at H=96, level at H=192, then 1.1% and 2.0% better at H=336 and H=720. Weather shows no such deficit, and the two differ in a way that predicts it: Weather enables the spectral pathway of Eq. (6) and Electricity does not, and the frequency head is zero-initialized *only* when that pathway is present (Section 4.2). Electricity therefore starts with a random frequency head, which we measured to inject ≈ 0.058 RMS of perturbation in the normalized space — essentially constant across horizons, and exactly zero when the pathway is on. A horizon-independent perturbation costs a larger share of a smaller error budget, matching the observed direction.

Enabling the pathway on Electricity failed informatively: it left H=336 unchanged but degraded H=720 by +0.024 (0.200 $\rightarrow$ 0.224), moving the dataset average from 0.169 to 0.175. The map interpolates the length-L spectrum onto a length-(L+H) grid, so its fan-out grows with the horizon — 49 $\rightarrow$ 97 bins at H=96 but 49 $\rightarrow$ 409 at H=720 — and the damage tracks that ratio; our configuration also applies no low-pass (ρ=0), whereas the ETT datasets using this pathway pair it with $\rho \in \{0.3, 0.4\}$, the cutoff that keeps linear-in-frequency forecasting stable. What the measurement implicates is the head's *initialization*, not the absence of a spectral map, and the head can be zeroed at no parameter cost and with no horizon-dependent extrapolation.

This is the leading explanation, not a demonstrated cause: the trend has four points, so any quantity monotone in H would correlate with it. Two further mechanisms point the same way and are not separated here — the architecture is linear-dominated, which favours long horizons approaching climatology, and the width-25 decomposition kernel smooths away sub-daily detail a 96-step forecast could use.

Complexity accounting.. Table 13 reports model-only estimates, not runtime evidence. Parameter count follows model *width*, not channel count (Electricity and Traffic both reach 19.7M despite a 2.7× gap in variates); compute and activation memory instead scale with channels through the variate mixer

(Traffic 60 GMACs vs. Electricity's 22 at equal width). That mixer is over 90% of parameters on high-channel data, so its cost is tunable: halving the width gives 5.3M, and weight-tying one mixer across branches gives 2.9M. On Traffic the tie is free (ΔMSE = +0.0007, CI $[-0.003, +0.005]$, three seeds), whereas a time-branch-only mixer is worse (+0.0045), a result PEMS04 reproduces (Table 11).

The reduction does *not* transfer (Table 12), and the two cases separate the mechanism. On Electricity, tying at full width is neutral as on Traffic; it is *halving the width* that costs +0.0045 average MSE ($0.1678 \to 0.1722$), enough to move DD-Mamba below the published S-Mamba average of 0.170 — with 321 channels and $\overline{|r|}$=0.50, the capacity that width controls is doing measurable work. On Weather the width is unchanged and merely *tying* the mixers costs +0.0013, concentrated at H=720: with 21 channels the mixer is small either way, but the branches evidently want *different* cross-channel mixing.

Sharing a mixer and shrinking one are therefore not interchangeable economies. We report the full-capacity configurations in Tables 3 and 4 and treat Table 12 as a measured accuracy–size trade-off rather than a free saving. Wall-clock throughput and matched-baseline runtime remain unmeasured.

6. Discussion

6.1. Data characteristics as validation hypotheses

The placement results suggest three hypotheses for future validation, not a deployment rule. First, stronger dependence between channels may justify testing a variate mixer (Fig. 4) — the PEMS04 result is the strongest evidence

Table 12: Measured cost of the parameter-reduced variants. "Shared" ties one variate mixer across the two branches; "d" is the model width. Δ is the change in MSE averaged over $H \in \{96, 192, 336, 720\}$ relative to the shipped configuration (positive is worse). Reductions that were free on Traffic are not free on Electricity or Weather.

Dataset	Configuration	Params	Avg MSE	Δ	Verdict
Traffic	d=256, shared	2.9M	—	+0.0007	neutral (CI incl. 0)
Electricity	d=512, shared	10.6M	0.1678	—	shipped
	d=256, shared	2.8M	0.1722	+0.0045	loses to S-Mamba
Weather	d=256, both	5.8M	0.2485	—	shipped
	d=256, shared	3.5M	0.2497	+0.0013	erases the margin

for this — but raw PCC ignores lagged and nonlinear structure. Second, a spectral path may help when the look-back does not resolve dominant periods; the present frequency evidence rests on a test-informed setting. Third, normalization may interact with changing scale and periodic zeros. Each choice must be selected on validation data and frozen before test evaluation.

6.2. Dispersion diagnostics: why normalization is dataset-dependent

A training-free analysis explores the two-sided RevIN result (Fig. 5). Standardizing each channel and taking a rolling standard deviation over the dominant period, all three probed datasets carry substantial time-varying scale (coefficient of variation of the rolling std 0.20–0.48); a KPSS test rejects stationarity of the rolling-std series in every sampled channel, and a Ljung-Box test finds it strongly autocorrelated. These tests describe nonstationary scale dynamics but do not prove forecastability or identify a RevIN mechanism. In the measured examples, removing normalization increases Traffic MSE by 0.100, decreases Solar MSE by 0.029, and decreases PEMS04

Table 13: Model-only complexity profile of DD-Mamba per dataset (look-back 96, horizon 96, batch 1). GMACs: multiply–accumulates per forecast (`Linear` + `Conv1d` + the selective-scan recurrence); act. mem.: forward activation footprint under a 32-bit representation. All three columns are hardware-independent estimates; no wall-clock or matched-baseline runtime is claimed.

Dataset	Variates	Params (M)	GMACs	Act. mem. (MB)
Exchange	8	0.09	0.02	2.3
ETTh1	7	0.25	0.08	4.0
Weather	21	5.77	1.90	46.6
Solar	137	0.96	3.26	150.4
Electricity	321	19.70	22.4	254.3
Traffic	862	19.70	60.2	682.7

MSE by 0.0038 (Table 11). Periodic night-time zeros are one possible explanation for Solar, but that explanation and the associated test-informed effects require validation on untouched data. Explicit dispersion forecasting in the style of STD [25] is outside the evaluated model.

6.3. Threats to validity

Several threats remain, each with the experiment that would resolve it.

Baseline scope and tuning.. The in-pipeline comparison covers six datasets and seven repo-native baselines, and one common schedule does not guarantee equal model-specific tuning. A broader ranking requires official implementations or verified adapters and comparable search budgets.

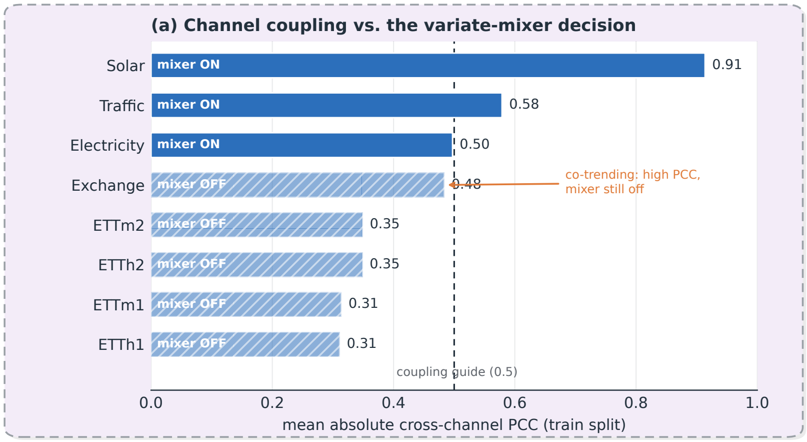

Figure 4: Channel coupling versus the variate-mixer decision. Bars give the mean absolute cross-channel Pearson correlation on the training split; solid bars are datasets where the mixer is enabled, hatched bars where it is disabled. Stronger coupling coincides with enabling the mixer, but Exchange is the instructive exception: its moderate correlation is co-trending rather than exploitable cross-sectional signal, so the mixer stays off. This is why we treat PCC as a screening *hypothesis*, not a selection rule. Weather is omitted (training-split PCC unavailable in this study).

The accuracy ceiling is generalization, not optimization.. Our schedule sets $\mathrm{lr} = \mathrm{base} \cdot 0.5^{(e-1)}$, a geometric series whose total budget is $2 \cdot \mathrm{base}$ *regardless of epoch count*, with only about four epochs above 10% of the base rate. Since the training loss is still falling at the final epoch, this looked like undertraining, so we tested it (`scripts/run_schedule_sweep.py`) by retraining under warmup-plus-cosine over thirty epochs, roughly five times the integrated rate. The hypothesis was wrong: the larger budget cut *training* loss by 16–31% across horizons while test error did not improve anywhere (dataset average $0.169 \to 0.171$), and the train-to-validation gap widened from 14–32% to

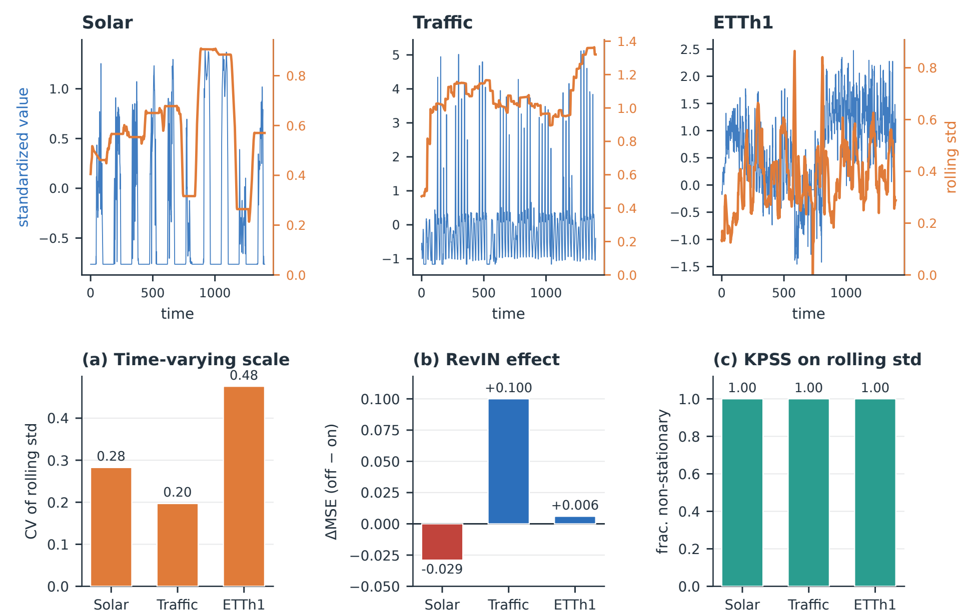

Figure 5: Training-free dispersion diagnostics. *Top:* a representative standardized channel (blue) with its rolling standard deviation (orange) for Solar, Traffic, and ETTh1. **(a)** Coefficient of variation of the rolling std. **(b)** Measured RevIN effect (ΔMSE, off−on): normalizing by the input scale helps Traffic but hurts Solar. **(c)** Fraction of sampled channels whose rolling-std series a KPSS test declares non-stationary. These summaries are descriptive; the Solar effect uses a test-informed configuration.

38–69%. The optimizer is not the constraint; the sharp decay was acting as an implicit regularizer.

What limits accuracy is the generalization gap: train-to-validation error rises 14–32% and validation-to-test a further 12–18% at short horizons, the latter reflecting distribution shift between the two periods. With reducing width also hurting (Table 12), the model sits near the useful capacity for this data and recipe. The sweep also strengthens the component-ablation nulls: the objection that zero-initialized components were never given budget to grow does not survive a five-fold larger one. The training loop emits an undertrained-versus-overfit diagnostic so the condition is visible per run.

Statistical power and multiplicity.. With n=3 seeds, tests have limited power, and although we apply Benjamini–Hochberg correction, all 29 component-placement effects fall to $q \geq 0.11$: those results are exploratory; five and preferably ten seeds are needed to move them from directional to confirmatory. Our pairing is also *positional* — seed index i of one configuration against index i of another — which shares the initialization seed but does not guarantee a verifiable common random-number stream across architectures with different parameter-sampling orders, so an unpaired or randomization-based sensitivity analysis should accompany the paired test.

Model-selection bias.. This is the most consequential threat. Checkpoints are chosen on validation loss, but some *switches* — most visibly disabling RevIN for Solar — were decided after inspecting test-set ablations. That is selection on the test signal, not training leakage, and it can inflate the affected datasets, so we exclude those findings from confirmatory claims and, for the same reason, decline the two PEMS04 changes that test-split ablations favour (Table 11). A clean follow-up fixes the candidate space in advance, selects every switch on validation data only, and reports a fixed, a validation-selected, and a test-oracle model.

Fusion-gate initialization bias.. The gate is biased to $g_0 \approx 0.9$, starting the model near the linear temporal pathway. This is an inductive bias, not a neutral prior: it down-weights the frequency and correction pathways during the highest-learning-rate epochs, and interacts with the init asymmetry of Section 4.2. Both could depress measured branch-removal effects, so the branch matrix is a lower bound on each domain's importance under this initialization; a gate-bias sweep and a matched random-vs-zero head-init comparison would quantify how much of the temporal dominance is architectural.

Channel-order sensitivity.. The bidirectional variate mixer is non-causal but not permutation-invariant, and all results use the canonical dataset column order, so part of the measured mixer effect — the largest effect in the paper — may depend on an arbitrary ordering. A permutation-equivariant mixer or repeated random-order evaluation is needed to resolve this.

Coverage and efficiency.. Electricity, Traffic, and Exchange enter the branch matrix at H=96 only, and the other component ablations cover only H=96; schedule sensitivity and head-norm dynamics are not measured; and the complexity table lacks wall-clock, matched-baseline throughput, latency, and peak-memory figures. The fixed 96-step look-back leaves the Traffic weekly-cycle hypothesis untested, while Exchange at H=720 remains a notable failure case. Explicit forecasting of time-varying dispersion is left as future work rather than claimed as part of the evaluated model.

7. Conclusion

DD-Mamba is a forecast-decomposable framework built on state-space branches — a DLinear-style temporal backbone, a selective-state-space correction, an optional frequency pathway, and a switchable cross-variate mixer, each an independently removable component. On the six datasets and seven baselines evaluated in one pipeline, it obtains the lowest average MSE on five datasets; corrected seed-matched tests support four differences, while Weather is tied and PatchTST is better on ETTm1. These results are restricted to the implemented comparison and do not establish a broader state-of-the-art ranking.

The component study provides a more qualified result. On the nine original benchmarks none of the 29 placement effects survives multiplicity correction at three seeds, while eight of 54 branch-removal tests support the temporal pathway, most consistently on ETTm1; the sole corrected frequency-side effect, Solar at H=96, is treated as exploratory because that dataset's normalization switch was selected with test feedback. PEMS04 separates components far more sharply: six of eleven effects survive, led by cross-variate mixing (+0.0232 MSE when removed) and reproducing S-Mamba's preference for a bidirectional selective scan over attention. Zero initialization is accuracy-neutral, and PCC and dispersion summaries remain descriptive covariates rather than validated selectors. The present evidence therefore supports forecast-decomposable analysis and dataset-specific hypotheses. A general component-selection rule requires validation-only configuration, untouched datasets, stronger baseline coverage, more ablation runs, channel permutation tests, and matched efficiency measurements.

CRediT authorship contribution statement

First Author: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Second Author:** Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All nine benchmark datasets are publicly available (ETT, Weather, Solar, Electricity, Traffic, Exchange). Code, configuration files, seed-level outputs, statistical-analysis scripts, and training logs will be deposited in an anonymized repository for review: *[repository URL to be inserted before submission]*.

Declaration of generative AI and AI-assisted technologies in the writing process

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